



# DANI CABALLERO MÉLICH

## AI ENGINEER · AGENTIC SYSTEMS

Barcelona · remote-first  
EU citizen, no sponsorship needed  
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### STACK

**Agents** LangGraph · MCP ·  
Pydantic · multi-agent  
orchestration · agent loops · tool  
use · human-in-the-loop gating ·  
guardrails

**Retrieval** RAG · hybrid dense +  
lexical · embeddings · sqlite-vec ·  
recall@k · MRR

**Evaluation** validator suites ·  
scored thresholds · output repair  
· observability

**Providers** Anthropic · OpenAI ·  
Claude Code · Codex CLI · per-  
stage model and effort routing ·  
fallback chains · cost and token  
telemetry

### PROGRAMMING

Python 3.12 · C# · Kotlin ·  
TypeScript · GDScript

### SYSTEMS

SQLite · JSON-RPC · asyncio ·  
pytest · Rust/Tauri · React · Git

### ALSO

Fully remote throughout,  
including cross-timezone work  
with a US studio.

Catalan and Spanish native ·  
English fluent.

I build multi-agent LLM pipelines: typed contracts, human gates, retrieval scored against a gold set, and a cost line on every stage. Shipping production software since 2023, on a product live on Google Play and the App Store.

### EXPERIENCE

#### Somni Game Studios · Software Engineer

2025 – 2026 · Stepland, live on Google Play and the App Store

- One of the core programming trio inside a larger team.
- Drove the coding-agent sessions that rebuilt the tutorial: 82 declarative steps, 76% of its surviving code, shipped to the 4YFN showcase date.
- Replaced a silently-failing third-party dependency with an in-house Kotlin plugin.
- Wrote it in a week, in a language I had not used; 211 of the sensor reader's 212 lines.
- Owned app boot, cloud sync and remote config. 7 of the 16 services are mine.

#### Independent · Software & AI Systems

2025 – present

- A 19-node LangGraph pipeline: 10 role-specialised agents, 5 human approval gates.
- Turns a written brief into validated Unity code, shipped into Dungeon Heroes.
- Typed Pydantic contracts are the only source of truth; every other view is generated.
- Evaluation harness: 9 validators plus scored visual comparison, gated at 0.85 / 0.70.
- Failures route back by conditional edge to the stage that caused them.
- Hybrid retrieval on a local sqlite-vec index, scored against a 32-pair gold set.
- Recall@10 0.97, MRR 0.66. Cost telemetry: \$0.95 per stage execution, 18 instrumented.
- ~15,300 lines of Python 3.12, 259 tests. SQLite checkpointing. Runs as an MCP server.

#### Purple Pwny Studios · Software Engineer

2023 – 2024 · remote, US studio

- Built an expert system — a heuristic policy under Monte Carlo simulation — as one reference point for automated economy balancing.
- Built a real-time Unity client end to end; contributed to the live-state web bridge.

### PROJECT

#### Dungeon Heroes · closed testing on Google Play

- The commercial mobile title the pipeline ships into. ~20,100 lines of C#.
- Versioned save-schema migrations, cloud sync, live-ops SDKs, signed release builds.

*Architecture write-up, telemetry and code walkthrough at [danimelich.com](https://danimelich.com)*